1. Evaluate the following integral:
$$\int_0^1 \int_{\sqrt{y}}^1 e^{x^3} \, dx \, dy.$$

2. Suppose a solid lies inside the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $z = \sqrt{x^2 + y^2}$. Its density is $\delta(x, y, z) = z^2$. Set up, but do not evaluate, a triple integral representing the mass of this solid in each coordinate system.

   (a) Cylindrical coordinates.

   (b) Spherical coordinates, with φ measured from the positive z-axis.

3. Suppose a solid lies above the paraboloid $z = x^2 + y^2$ and below the paraboloid $z = 8 - x^2 - y^2$. Set up a triple integral in an appropriate coordinate system representing the volume of this solid, and evaluate it.

4. A wire follows the curve parametrized by
$$\vec{r}(t) = (t^2, 3t, e^t), \qquad 0 \le t \le 1,$$
and has density $\delta(x, y, z) = y + z$. Set up, but do not evaluate, a line integral representing the mass of the wire.

5. Compute the work done by the force field $\vec{F}(x, y) = (-y, x)$ on a particle moving once around the ellipse
$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$
in the **clockwise** direction.

6. Consider the vector field
$$\vec{F}(x, y) = \left(ye^x + 2x, \; e^x + 2y\right)$$
and the curve C along the parabola $y = x^2 + 1$, oriented from $(0, 1)$ to $(1, 2)$.

   (a) Determine whether $\vec{F}$ is conservative.

   (b) If $\vec{F}$ is conservative, find a potential function f. Then use the Fundamental Theorem of Line Integrals to evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$.